UF1. Introduction to HTML5

"Elements inside

the <body> section"



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1. REVIEW…

- Basic structure of an HTML5 document

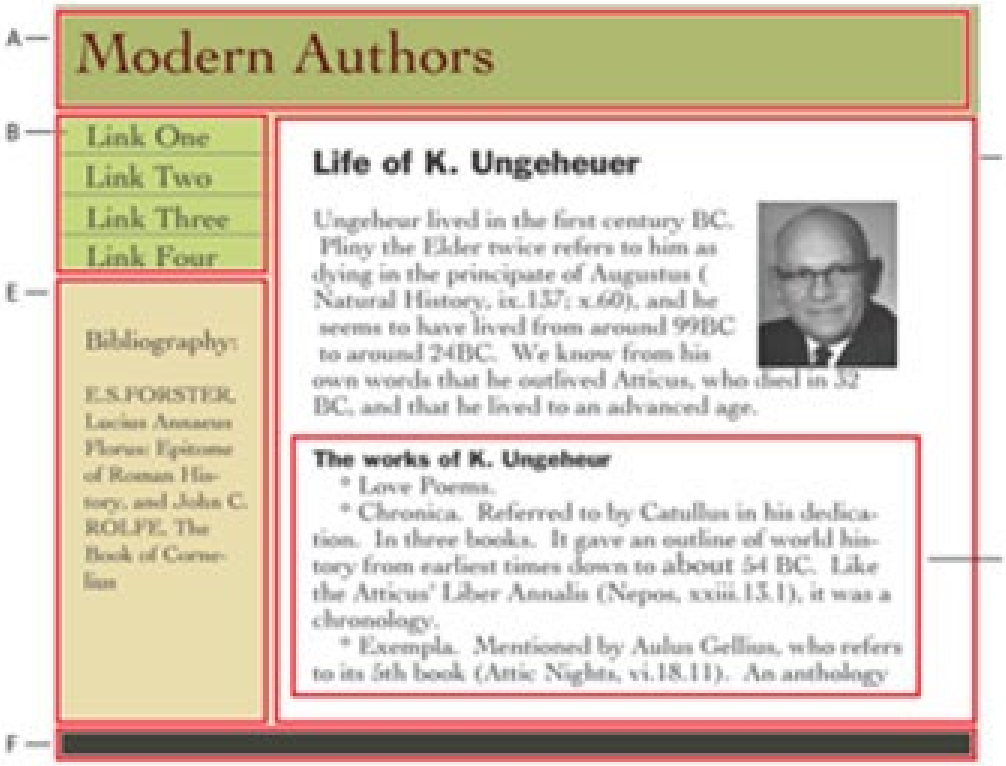
```
<!DOCTYPE html>
<html lang="en">
 <head>
    ... header of html document ...
 </head>
 <body>
    ... body of  html document (content of the webpage) ...
 </body>
</html>
```

1. REVIEW…

- Semantic containers inside the body:

```
<!doctype html>
<html>
   <head>
      <title>T&iacute;tulo de P&aacute;gina</title>
   </head>
   <body>
      <header>
         <h1>T&iacute;tulo de P&aacute;gina</h1>
      </header>
      <nav>
         <!-- Navegación -->
      </nav>
      <section id="intro">
         <!-- Introducción -->
      </section>
      <section id="intro">
         <!-- Contenido principal
      </section>
      <aside>
         <!-- Barra Lateral -->
      </aside>
      <footer>
         <!-- Pie de Página -->
      </footer>
   </body>
</html>
```

+ info: https://11bits.es/etiquetas-estructurales-en-html5/



2. Grouping of contents

In HTML5 the basic elements, as well the basic containers for textual content, are the same as HTML 4.01

- **<h1>..</h1> <h2>..</h2> …. <h6>..</h6>**

  <h1> Title </h1> : tags for 1st level title-header
- ** ... **: paragraph
- ** … **: block divisor (multipurpose container, used as 'section' in HTML4)
- **…**: in-line grouping of elements

Other useful elements:

- **
**: line-break
- **<hr />**: horizontal row
- **<!-- comments -->**: special pseudo-tag to make comments

  <!-- text inside is not showed by the browser -->

3. Textual content

- Textual content **must** be placed inside a container:

  texto <h2>texto</h2> texto

- Special characters, as **á, é, í, ó, ú, ñ**,… initially had to be codified with pseudo-tags:

Caracter	Códigos	Caracter	Códigos	Caracter	Códigos
á	á	Ú	Ú	<	<
Á	Á	Ü	ü	>	>
é	é	Ü	Ü	&	&
É	É	Ñ	ñ	"	"
í	í	Ñ	Ñ	º	º

- However, now is not necessary, as long as you declare a <meta> tag in the header, that declares that the page uses a extended character set:



<meta charset="utf-8" />

<meta charset="iso-8859-1" /> <!-- ISO which defines the codification of Latin alphabet -->

- Browsers tend to minimize space characters, so…

  Good morning is shown as "Good morning" *(one space)*

- If we want more than one space, we use pseudo-tag ** **

  Good   morning

3.1 Semantic mark-up of textual content



- One of the goals of HTML5 is to improve the semantic definition of page elements
- So, we have different semantic elements, in order to give sense and structure, to all the content included in our page.
  Note: Browser can have predefined stiles to show these elements, but we can make our own styles by using CSS3:
  - Important words .. => **bolt** print, by default
  - Emphasize .. => **italic**, by default
  - Highlight <mark>..</mark> => background in **yellow**, by default
  - Cite <cite>..</cite> => **italic**, by default. P.e. when we cite someone else
  - Reduce <small>..</small> => **small** size, p.e. a note at the footer of page
  - Code <code>..<code> => **monospaced** font (courier), p.e. source code samples
  - Cross out ~~..~~ => **overcrossed**, p.e. information out-of-date or invalid
  - Definition and abbreviate <dfn>... </dfn> <abbr>…</abbr>
  - Text inserted <ins> …</ins>

  + info: https://developer.mozilla.org/es/docs/Web/HTML/Element

3.1 Semantic mark-up of textual content

Visual mark-up of contents:

- We have tags that are to visually highlight some content. These tags are not recommended. It is better to use the semantic tags instead:
  - <b>: bold => better use
  - <i> : italic => better use
  - <s>: stroke line => better use <del>
  - <u>: underline => better do it with CSS





- It is recommended using **semantic mark-up** (as shown in previous page) The semantic tags will improve the SEO* of the our webpage
- If we want to give visual design to any paragraph or word, better use CSS3

(*): This is useful for SEO, though also we could use schema.org for better web positioning
+info: http://blogs.icemd.com/blog-marketing-en-buscadores-seo-ppc-y-analitica-web-/como-usar-el-marcado-semantico-de-schema-org/

4. Lists

- Two kind of lists :
  - Ordered lists: ** … **
  - Unordered: **
… **

- Each item element inside a list, specified with <li>.. tags

  ** Element 1   Element 2 …**

- **Attributes of <ol> :**

  Better use CSS properties

  list-style-type

  - start="num": Sets the first number on the list (1 a default)
  - type="tipo": Sets the type of numbering DEPRECATED
    - "1" ⇨ numeric (1,2,3,4...)
    - "a" ⇨ lowercase alphabetic (a,b,c,d...)
    - "A" ⇨ Uppercase alphabetic (A,B,C,D...)
    - "i" ⇨ lowercase Roman numbers (i,ii,iii,iv...)
    - "I" ⇨ Uppercase Roman numbers (I,II,III,IV...)



- **Attributes of <ul> :**
  - "disc" •
  - "circle" °
  - "square" □



Better use CSS properties

list-style-type

4. Lists

Nested lists, we have sublevels…

Example:

```
<ol>
   <li> First item</li>
   <li> Second item
        <ul>
             <li> first nested item </li>
             <li> second nested item </li>
        </ul>
   </li>     <!-- We close after the sublist-->
   <li> Third item </li>
</ol>
```

5. Tables

- **<table>...</table>**
- **Attributes:**
  - width="pixels" or "%"
  - height="pixels" or "%"

  Other visual attributes are deprecated (we must use CSS3 instead)
  - ~~border="pt": table border~~
  - ~~cellspacing="pt": space between rows~~
  - ~~cellpadding="pt": space from border to the inside content of the cell~~
  - ~~bgcolor="color": background color of the table~~



- Inside the table element, we use
  - **<tr> … </tr>** for each of the Rows
  - And **<td> … </td>** for each of the Cell inside the Row
    (In 1st row, if cells are headers we could use **<th>…</th>**)
- Less used elements are **<thead>**, **<tbody>**, **<tfoot>** and **<caption>**

5. Tables

- We can expand the cells **td**, by using attributes **colspan** y **rowspan**

```
<table width="80%">
  <tr>
    <td colspan="3">Cell expanded horizontally</td>
  </tr>
  <tr>
    <td rowspan="2" width="50%">Cell exp. vertically</td>
    <td>data</td>
    <td>data</td>
  </tr>
  <tr>
    <td>data</td>
    <td>data</td>
  </tr>
</table>
```

Better use CSS properties

column-span

- Not allowed other visual attributes, as: ~~cellspacing, cellpadding, bgcolor, align, valign, border, hspace, vspace~~… We'll use CSS



5.1. Page Layout using Tables

Historically, some Web authors have used tables as a way to control their page layout.



```
<!DOCTYPE HTML>
<html>
<head> … </head>
<body>
<table width="100%">
 <tr>
    <td colspan="2">
       <header> … </header>
    </td>
 </tr>
 <tr>
    <td>
        <nav> … </nav>
    </td>
    <td>
        <section id="principal"> … </section>
    </td>
 </tr>
 <tr>
    <td colspan="2">
       <footer> … </footer>
    </td>
  </tr>
</table>
</body>
</html>
```

NOT RECOMMENDED!

Tables should not be used for page layout! There are a lots of CSS alternatives for page layout.

5.1. Page Layout using Tables

Reasons for not using tables for page layout…

- Table element is for **tabulating information**
  - That's its main objective
- We must **keep a part CONTENT from Visual layout**
  - We use HTML to structure semantically the content
  - Latter we use CSS to design the visual layout
- We have the **power of CSS:** We have a lot of different techniques in CSS3, in order to manage the page layout
  - We'll go through them in another Unit

However, in order to give layout to small parts of the content, could be useful sometimes:

```
<table>
  <tr>
     <td><img src="img1.jpg"/></td>
    <td><p>A explanation of the image...</p></td>
  </tr>
</table>
```

6. Images

- To insert images in our page, we use element ****.
- With attribute **src**, we specify the image location:

  ****

- **Attributes:**
  - **src="url"**: source location* of the file on the server (*relative or abs)
  - **alt="texto"**: alternative text (useful for accessibility)
  - **width="100"**: width of image (number of pixels or %) *
  - **height="100"**: height of image (pixels or %) *

    width & height have to be declared in CSS, but it is interesting dectare them in HTML "for pre-rendering" reasons
  - **usemap="mapoflinks"**: we can define images with hyperlinks**

    "usemap" attribute will be introduced when we'll see multimedia features

- Not allowed use other HTML4 attributes as: ~~align, border, hspace, vspace~~… We will use CSS3



6.1 Locations of the files

Ways to specify location of files (or directories):

1. **Absolute path** to a file location: points the file location, starting from the root of the files system:

   p. e. **C:/Program Files/Xamp/htdocs/img/img1.jpg**
   /usr/apache/www/prj1/img/img2.jpg

2. **Relative path**: points the file location from the current location (this is the location where .html document is saved on the server):

   p. e. **./dir1/arc1.html** is pointing arc1.html file, which is inside dir1 folder of the current location

   dir1/arc1.html this is equivalent (so, we don't really need to write "./")

   ../dir1/arc1.html two dots are pointing the PARENT folder of where we are now

Considerations:

- In HTML, we always must use **relative paths**, so when we upload our project from local to the server, we don't have path reference problems.
- Regarding the names of folders and files :
  - Avoid accents, spaces between words, "ñ" and other special chars
  - Be aware of using UPPERCASE in some file and folder names
    -> better if we use only lowercase.

7. Hyperlinks

- The use of hyperlinks in HTML documents, is a key in the succees of the WWW
- The element **<a> … ** (very easy) converts everything (text, image, etc) into a link

  ****

  From then, this image will be a hyperlink, and the mouse reacts when dragging over it
- A hyperlink could point to…
  - …a section in the same page: **Inicio de la sección1**
  - …other page/doc in a local path: **Descargar apuntes**
    Ir a página 2
  - …other page/doc that are in a remote server:

    Ir a google
  - …reference to other Internet Services:

    Escribe tus sugerencias

- **Attributes:**
  - **href=“url”**: specifies the URL of resource
  - **target=“_blank”**: destination will be opened in a new browser window. Other: _self, framename
  - ~~**name=“seccion1”**: DEPRECATED. In HTML4, was used to create “anchors” inside our doc.~~

7. Hyperlinks

"Linking" to "anchors" inside the same webpage

- 1. We can use attribute **id** in any container (it will work as a anchor):

  <section id="section2" > … </section>

- 2. When declaring **<a>** tag, href must point to id of section, preceded by # character

  **Go to section 2 **

  We don't need to declare an "anchor" that points to the begging of the de document. We just point it declaring "#" as href :

  Go to top

"Linking" to other "local" webpages (allocated in the same server)

- When declaring **<a>** tag, href must point to relative path of the page:

  **Got to page 2 **

- We can also point to an "anchor" of the other page:

  **Got to page2, sect.2 **

7. Hyperlinks

"Linking" to external resources (allocated in a different server)

```
<a href="http://www.ua.es">Universidad de Alicante </a>
```

- When clicking on the hyperlink, the current page will be replaced by the referenced
- We can use attribute **target="_blank"**, to open the referenced resource in a new browser window:

```
<a href="http://www.upv.es" target="_blank">U.P.V. </a>
```

- We can also use **target="*frame_name*"**: Resource will be open in specified iframe.
  This feature will be introduced when we'll see iframe , when reviewing multimedia

"Link" which allows e-mail composing

- In this case, in href, we will specify the tag "mailto" and the address we want to contact

```
<a href="mailto: support@opp.es">Send an email to support </a>
```

This hyperlink will open the default email client installed in the computer (Outlook, Firebird) or GMAIL. Then will display a new blank email, but with the "To:" field already filled.

"Link" which allows dialing

```
<a href="tel:+34912223344">Call us </a>
```

This hyperlink will open the telephone dialler installed, with the tel. number written.